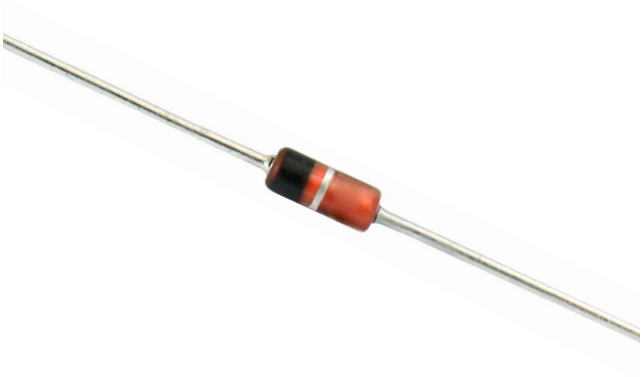
Small Signal Zener Diodes



FEATURES

- Very sharp reverse characteristic
- Low reverse current level
- Very high stability
- Low noise
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

APPLICATIONS

- Voltage stabilization

LINKS TO ADDITIONAL RESOURCES

3D Models | Marking | Parametric Search | Order Samples

PRIMARY CHARACTERISTICS		
PARAMETER	VALUE	UNIT
V_Z range nom.	2.4 to 36	V
Test current I_{ZT}	2; 5	mA
V_Z specification	Pulse current	
Circuit configuration	Single	

ORDERING INFORMATION			
DEVICE NAME	ORDERING CODE	TAPED UNITS PER REEL	MINIMUM ORDER QUANTITY
TZX-series	TZX-series-TAP	10 000 per ammopack (52 mm tape)	30 000/box
TZX-series	TZX-series-TR	10 000 per 14" reel (52 mm tape)	30 000/box

PACKAGE				
PACKAGE NAME	WEIGHT	MOLDING COMPOUND FLAMMABILITY RATING	MOISTURE SENSITIVITY LEVEL	SOLDERING CONDITIONS
DO-35 (DO-204AH)	approx. 125 mg	UL 94 V-0	MSL level 1 (according J-STD-020)	Peak temperature max. 260 °C

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Power dissipation	l = 4 mm, T_L = 25 °C	P_{tot}	500	mW
Zener current		I_Z	P_{tot}/V_Z	mA
Thermal resistance junction to ambient air	l = 4 mm, T_L =constant	R_{thJA}	300	K/W
Junction temperature		T_j	175	°C
Storage temperature range		T_{stg}	-65 to +175	°C
Forward voltage (max.)	I_F = 200 mA	V_F	1.5	V

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

PART NUMBER	ZENER VOLTAGE RANGE		TEST CURRENT	REVERSE LEAKAGE CURRENT				DYNAMIC RESISTANCE
	V_Z at I_{ZT1}		I_{ZT1}	I_R at V_R		I_R at V_R (1)		Z_Z at I_{ZT1}
	V		mA	µA	V	µA	V	Ω
	MIN.	MAX.		MAX.		MAX.		MAX.
TZX2V4A	2.3	2.5	5	5	0.5	50	1	100
TZX2V4B	2.4	2.6	5	5	0.5	50	1	100
TZX2V7A	2.5	2.7	5	5	0.5	10	1	100
TZX2V7B	2.6	2.8	5	5	0.5	10	1	100
TZX2V7C	2.7	2.9	5	5	0.5	10	1	100
TZX3V0A	2.8	3	5	5	0.5	6	1	100
TZX3V0B	2.9	3.1	5	5	0.5	6	1	100
TZX3V0C	3	3.2	5	5	0.5	6	1	100
TZX3V3A	3.1	3.3	5	5	1	2	1	100
TZX3V3B	3.2	3.4	5	5	1	2	1	100
TZX3V3C	3.3	3.5	5	5	1	2	1	100
TZX3V6A	3.4	3.6	5	5	1	2	1	100
TZX3V6B	3.5	3.7	5	5	1	2	1	100
TZX3V6C	3.6	3.8	5	5	1	2	1	100
TZX3V9A	3.7	3.9	5	5	1	2	1	100
TZX3V9B	3.8	4	5	5	1	2	1	100
TZX3V9C	3.9	4.1	5	5	1	2	1	100
TZX4V3A	4	4.2	5	5	1.5	1	1	100
TZX4V3B	4.1	4.3	5	5	1.5	1	1	100
TZX4V3C	4.2	4.4	5	5	1.5	1	1	100
TZX4V3D	4.3	4.5	5	5	1.5	1	1	100
TZX4V7A	4.4	4.6	5	5	2	6	2	100
TZX4V7B	4.5	4.7	5	5	2	5	2	100
TZX4V7C	4.6	4.8	5	5	2	4	2	100
TZX4V7D	4.7	4.9	5	5	2	3	2	100
TZX5V1A	4.8	5	5	5	2	2	2	100
TZX5V1B	4.9	5.1	5	5	2	2	2	100
TZX5V1C	5	5.2	5	5	2	2	2	100
TZX5V1D	5.1	5.3	5	5	2	2	2	100
TZX5V6A	5.2	5.5	5	5	2	1	2	40
TZX5V6B	5.3	5.6	5	5	2	1	2	40
TZX5V6C	5.4	5.7	5	5	2	1	2	40
TZX5V6D	5.5	5.8	5	5	2	1	2	40
TZX5V6E	5.6	5.9	5	5	2	1	2	40
TZX6V2A	5.7	6	5	1	3	3	4	15
TZX6V2B	5.8	6.1	5	1	3	3	4	15
TZX6V2C	6	6.3	5	1	3	3	4	15
TZX6V2D	6.1	6.4	5	1	3	3	4	15
TZX6V2E	6.3	6.6	5	1	3	3	4	15
TZX6V8A	6.4	6.7	5	1	3.5	2	4	15
TZX6V8B	6.6	6.9	5	1	3.5	2	4	15
TZX6V8C	6.7	7	5	1	3.5	2	4	15
TZX6V8D	6.9	7.2	5	1	3.5	2	4	15

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

PART NUMBER	ZENER VOLTAGE RANGE		TEST CURRENT	REVERSE LEAKAGE CURRENT				DYNAMIC RESISTANCE
	V_Z at I_{ZT1}		I_{ZT1}	I_R at V_R		I_R at V_R (1)		Z_Z at I_{ZT1}
	V		mA	µA	V	µA	V	Ω
	MIN.	MAX.		MAX.		MAX.		MAX.
TZX7V5A	7	7.3	5	1	5	30	6.65	15
TZX7V5B	7.2	7.6	5	1	5	30	6.84	15
TZX7V5C	7.3	7.7	5	1	5	30	6.94	15
TZX7V5D	7.5	7.9	5	1	5	30	7.13	15
TZX7V5X	7.07	7.45	5	1	5	30	6.72	15
TZX8V2A	7.7	8.1	5	1	6.2	0.1	7.32	20
TZX8V2B	7.9	8.3	5	1	6.2	0.1	7.5	20
TZX8V2C	8.1	8.5	5	1	6.2	0.1	7.7	20
TZX8V2D	8.3	8.7	5	1	6.2	0.1	7.98	20
TZX9V1A	8.5	8.9	5	1	6.8	0.04	8.08	20
TZX9V1B	8.7	9.1	5	1	6.8	0.04	8.27	20
TZX9V1C	8.9	9.3	5	1	6.8	0.04	8.46	20
TZX9V1D	9.1	9.5	5	1	6.8	0.04	8.65	20
TZX9V1E	9.3	9.7	5	1	6.8	0.04	8.84	20
TZX10A	9.5	9.9	5	1	7.5	0.04	9.03	25
TZX10B	9.7	10.1	5	1	7.5	0.04	9.22	25
TZX10C	9.9	10.3	5	1	7.5	0.04	9.41	25
TZX10D	10.2	10.6	5	1	7.5	0.04	9.69	25
TZX11A	10.4	10.8	5	1	8.2	0.04	9.88	25
TZX11B	10.7	11.1	5	1	8.2	0.04	10.2	25
TZX11C	10.9	11.3	5	1	8.2	0.04	10.4	25
TZX11D	11.1	11.6	5	1	8.2	0.04	10.5	25
TZX12A	11.4	11.9	5	1	9.5	0.04	10.8	35
TZX12B	11.6	12.1	5	1	9.5	0.04	11	35
TZX12C	11.9	12.4	5	1	9.5	0.04	11.3	35
TZX12D	12.2	12.7	5	1	9.5	0.04	11.6	35
TZX12X	11.44	12.03	5	1	9.5	0.04	10.9	35
TZX13A	12.4	12.9	5	1	10	0.04	11.8	35
TZX13B	12.6	13.1	5	1	10	0.04	12	35
TZX13C	12.9	13.4	5	1	10	0.04	12.3	35
TZX14A	13.2	13.7	5	1	11	0.04	12.5	35
TZX14B	13.5	14	5	1	11	0.04	12.8	35
TZX14C	13.8	14.3	5	1	11	0.04	13.1	35
TZX15A	14.1	14.7	5	1	11.5	0.04	13.4	40
TZX15B	14.5	15.1	5	1	11.5	0.04	13.8	40
TZX15C	14.9	15.5	5	1	11.5	0.04	14.2	40
TZX15X	14.35	15.09	5	1	11.5	0.04	13.6	40
TZX16A	15.3	15.9	5	1	12	0.04	14.5	45
TZX16B	15.7	16.5	5	1	12	0.04	14.9	45
TZX16C	16.3	17.1	5	1	12	0.04	15.5	45
TZX18A	16.9	17.7	5	1	13	0.04	16.1	55
TZX18B	17.5	18.3	5	1	13	0.04	16.6	55
TZX18C	18.1	19	5	1	13	0.04	17.2	55
TZX20A	18.8	19.7	2	1	15	0.04	17.9	60
TZX20B	19.5	20.4	2	1	15	0.04	18.5	60
TZX20C	20.2	21.2	2	1	15	0.04	19.2	60
TZX22A	20.9	21.9	2	1	17	0.04	19.9	65

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)								
PART NUMBER	**ZENER VOLTAGE RANGE**		**TEST CURRENT**	**REVERSE LEAKAGE CURRENT**				**DYNAMIC RESISTANCE**
	V_Z at I_{ZT1}		I_{ZT1}	I_R at V_R		I_R at V_R (1)		Z_Z at I_{ZT1}
	V		mA	µA	V	µA	V	Ω
	MIN.	MAX.		MAX.		MAX.		MAX.
TZX22B	21.6	22.6	2	1	17	0.04	20.5	65
TZX22C	22.3	23.3	2	1	17	0.04	21.2	65
TZX24A	22.9	24	2	1	19	0.04	21.8	70
TZX24B	23.6	24.7	2	1	19	0.04	22.4	70
TZX24C	24.3	25.5	2	1	19	0.04	23.1	70
TZX24X	22.61	23.77	2	1	19	0.04	21.5	70
TZX27A	25.2	26.6	2	1	21	0.04	23.9	80
TZX27B	26.2	27.6	2	1	21	0.04	24.9	80
TZX27C	27.2	28.6	2	1	21	0.04	25.8	80
TZX27X	26.99	28.39	2	1	21	0.04	25.6	80
TZX30A	28.2	29.6	2	1	23	0.04	26.8	100
TZX30B	29.2	30.6	2	1	23	0.04	27.7	100
TZX30C	30.2	31.6	2	1	23	0.04	28.7	100
TZX30X	29.02	30.51	2	1	23	0.04	27.6	100
TZX33A	31.2	32.6	2	1	25	0.04	29.6	120
TZX33B	32.2	33.6	2	1	25	0.04	30.6	120
TZX33C	33.2	34.5	2	1	25	0.04	31.5	120
TZX36A	34.2	35.7	2	1	27	0.04	32.5	140
TZX36B	35.3	36.8	2	1	27	0.04	33.5	140
TZX36C	36.4	38	2	1	27	0.04	34.6	140
TZX36X	35.36	37.19	2	1	27	0.04	33.6	140

Notes

- Additional measurement of voltage group TZM9V1 to TZX36, I_R at 95 % $V_{Zmin.}$ ≤ 40 nA at T_j = 25 °C

(1) Additional measurement

BASIC CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

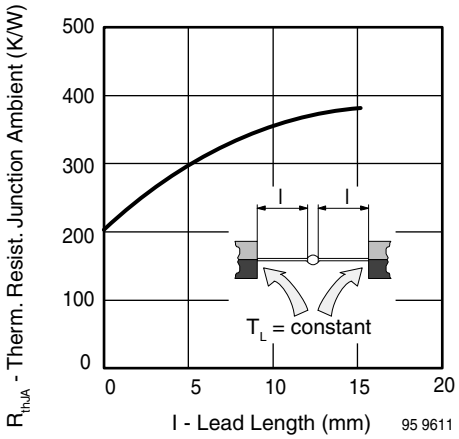


Fig. 1 - Thermal Resistance vs. Lead Length

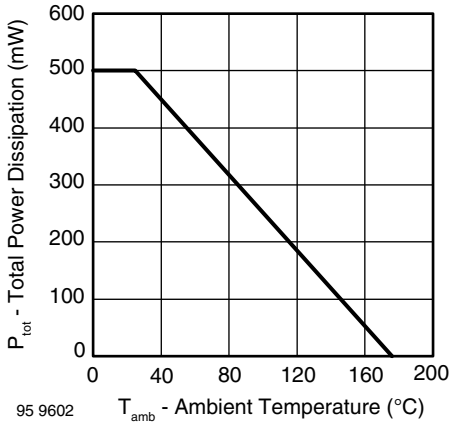


Fig. 2 - Total Power Dissipation vs. Ambient Temperature

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

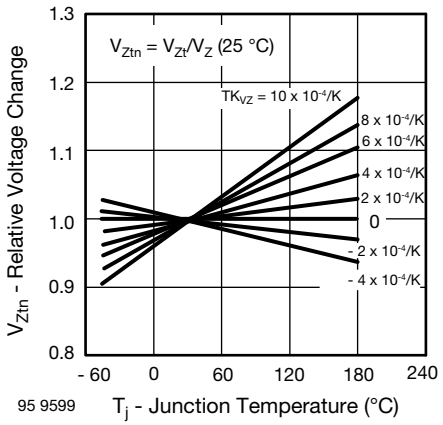


Fig. 3 - Typical Change of Working Voltage vs. Junction Temperature

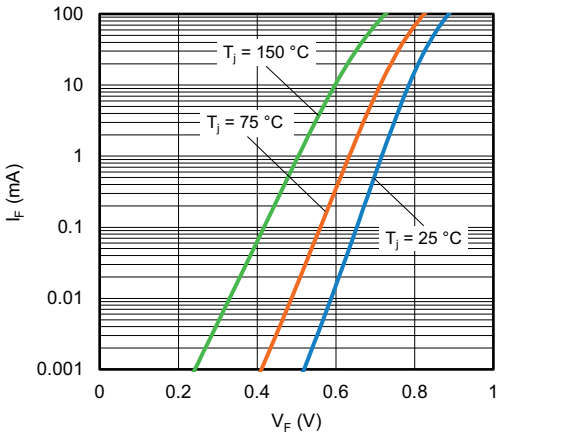


Fig. 6 - Typical Forward Current I_F vs. Forward Voltage V_F

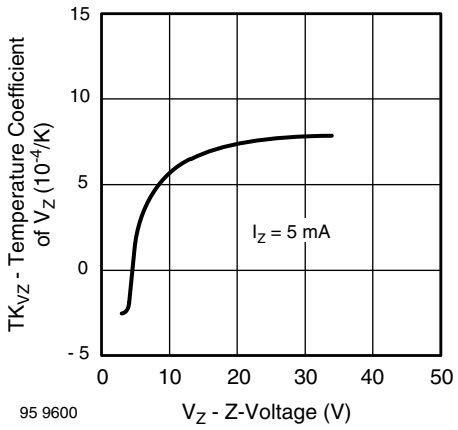


Fig. 4 - Typical Temperature Coefficient of Vz vs. Z-Voltage

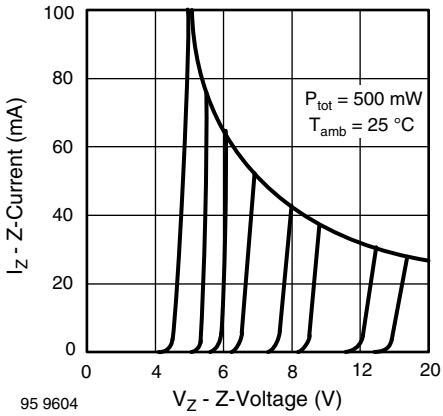


Fig. 7 - Typical Z-Current vs. Z-Voltage

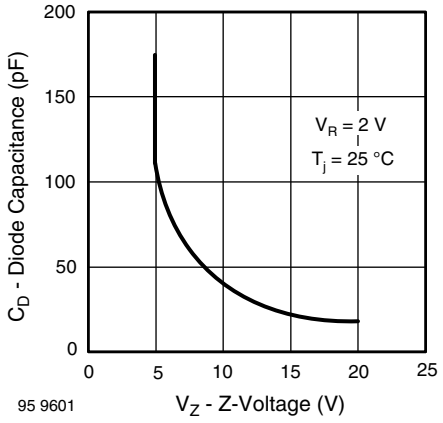


Fig. 5 - Diode Capacitance vs. Z-Voltage

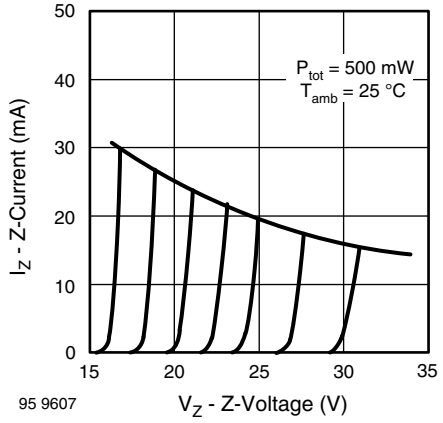


Fig. 8 - Typical Z-Current vs. Z-Voltage

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

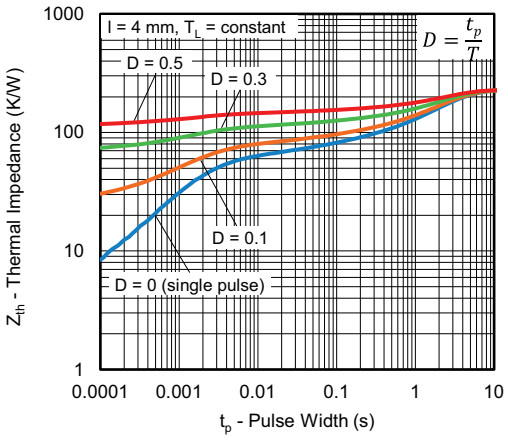
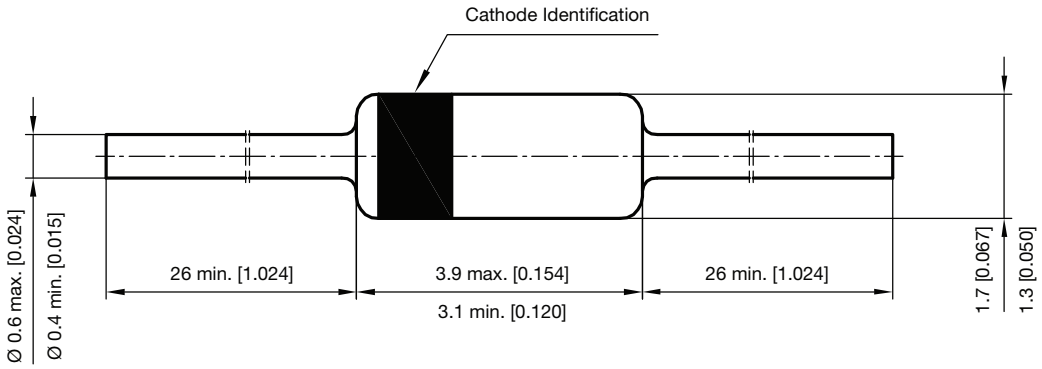


Fig. 9 - Typical Thermal Response

PACKAGE DIMENSIONS in millimeters (inches): **DO-35 (DO-204AH)**



Rev. 6 - Date: 19. December 2011
Document no.: SB-V-3906.04-031(4)
94 9366

For technical questions within your region: DiodesAmericas@vishay.com, DiodesAsia@vishay.com, DiodesEurope@vishay.com
THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000

Disclaimer

ALL PRODUCT, PRODUCT SPECIFICATIONS AND DATA ARE SUBJECT TO CHANGE WITHOUT NOTICE TO IMPROVE RELIABILITY, FUNCTION OR DESIGN OR OTHERWISE.

Vishay Intertechnology, Inc., its affiliates, agents, and employees, and all persons acting on its or their behalf (collectively, “Vishay”), disclaim any and all liability for any errors, inaccuracies or incompleteness contained in any datasheet or in any other disclosure relating to any product.

Vishay makes no warranty, representation or guarantee regarding the suitability of the products for any particular purpose or the continuing production of any product. To the maximum extent permitted by applicable law, Vishay disclaims (i) any and all liability arising out of the application or use of any product, (ii) any and all liability, including without limitation special, consequential or incidental damages, and (iii) any and all implied warranties, including warranties of fitness for particular purpose, non-infringement and merchantability.

Statements regarding the suitability of products for certain types of applications are based on Vishay's knowledge of typical requirements that are often placed on Vishay products in generic applications. Such statements are not binding statements about the suitability of products for a particular application. It is the customer's responsibility to validate that a particular product with the properties described in the product specification is suitable for use in a particular application. Parameters provided in datasheets and / or specifications may vary in different applications and performance may vary over time. All operating parameters, including typical parameters, must be validated for each customer application by the customer's technical experts. Product specifications do not expand or otherwise modify Vishay's terms and conditions of purchase, including but not limited to the warranty expressed therein.

Hyperlinks included in this datasheet may direct users to third-party websites. These links are provided as a convenience and for informational purposes only. Inclusion of these hyperlinks does not constitute an endorsement or an approval by Vishay of any of the products, services or opinions of the corporation, organization or individual associated with the third-party website. Vishay disclaims any and all liability and bears no responsibility for the accuracy, legality or content of the third-party website or for that of subsequent links.

Vishay products are not designed for use in life-saving or life-sustaining applications or any application in which the failure of the Vishay product could result in personal injury or death unless specifically qualified in writing by Vishay. Customers using or selling Vishay products not expressly indicated for use in such applications do so at their own risk. Please contact authorized Vishay personnel to obtain written terms and conditions regarding products designed for such applications.

No license, express or implied, by estoppel or otherwise, to any intellectual property rights is granted by this document or by any conduct of Vishay. Product names and markings noted herein may be trademarks of their respective owners.

© 2026 VISHAY INTERTECHNOLOGY, INC. ALL RIGHTS RESERVED

THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE. THE PRODUCTS DESCRIBED HEREIN AND THIS DOCUMENT ARE SUBJECT TO SPECIFIC DISCLAIMERS, SET FORTH AT www.vishay.com/doc?91000